

STATEMENT OF WORK/REQUIREMENTS DOCUMENT

Title: Acquisition of IMU and EMG System

Requesting Lab: Engineering Lab (EL),

I. BACKGROUND INFORMATION

The item being acquired is a sensing system with two parts: an IMU (Inertial Measurement unit)-based whole-body kinematic tracking system, and an EMG (Electromyography) measurement system. The IMUs are a set of sensors that can be worn by a user, and the pose of the body over time can be measured. The EMG system measures muscle effort through passive electrical signals in the skin. This sensing system will be used by the Mobility Performance of Robotic Systems project to measure the physical state of users while they operate exoskeletons. The system will be used for data collection in the lab and prepared locations outside the lab. It will be used in conjunction with previously purchased exoskeleton devices, but is not directly connected to any of them, it is solely for evaluating the user. Generally, it is required that the system provide the sensing capabilities needed for the project, be able to be used together with a variety of exoskeleton devices, and be compatible with the pre-existing sensing systems used on the project.

Purpose/Objective: The purpose of these requirements are to detail the needs of the Mobility Performance of Robotic Systems for a sensing system to aid in human observation while operating exoskeletons.

II. SCOPE

The Contractor shall deliver a quantity of one (1) IMU-based human pose measurement system, one (1) EMG measurement system, and at least one (1) license to a corresponding software through which data can be collected from both measurement systems and processed. These items are inclusive of FOB Destination delivery, installation, warranty and training, and option line items.

III. MINIMUM REQUIREMENTS

The Contractor shall provide a system that meets all technical specifications identified below. All items must be new. Used or remanufactured equipment will not be considered for award. Experimental, prototype, or custom items will not be considered. The use of “gray market” components are not authorized for sale in the U.S. by the Contractor is not acceptable. All line items shall be shipped in the original manufacturer’s packaging and include all original documentation and software, when applicable.

Line Item 0001: IMU-based human pose measurement system

Description: The system shall consist of a set of sensors sufficient for full-body motion tracking of one (1) user. The sensors must be able to be worn at the same time as an exoskeleton device. All accessories required for the charging of, communication with, and wearing of these sensors shall be provided with the system.

Quantity: One (1) system, consisting of multiple sensors

A. Technical Specifications

- a. The IMU-system must be able to measure rigid body positions with a static angular accuracy of less than 2 degrees.
- b. Each IMU device must have a battery life of at least 4 hours.

- c. The devices must be capable of recording data at a rate of at least 100hz.
- d. The system must be able to stream data from all sensors to an in-room computer for storage and processing. **(no cloud processing due to government restrictions)**

Line Item 0002: EMG measurement system

Description: The system shall consist of a set of sensors capable of measuring EMG signals at least eight (8) different locations on the body. The sensors must be able to be worn at the same time as an exoskeleton device. All accessories required for the charging of, communication with, and wearing of these sensors shall be provided with the system.

Quantity: One (1) system, consisting of multiple sensors

A. Technical Specifications

- a. The system must be sensitive enough to identify instances and relative magnitude of muscle effort (generally less than 2 μ V).
- b. Each IMU device must have a battery life of at least 4 hours.
- c. The devices must be capable of recording data at a rate of at least 100hz.
- d. The system must be able to stream data from all sensors to an in-room computer for storage and processing. **(no cloud processing due to government restrictions)**

Line Item 0003: Software License

Description: The delivered software must be capable of communicating with all sensors of both the EMG and IMU systems simultaneously. It must be able to, at minimum, record the raw data and perform preliminary processing to prepare it to be exported. It must be able to synchronize all collected data, and have the capacity to set up time-synchronization with third-party devices. All accessories required to successfully use the software with the sensing systems on a pre-existing computer must be included.

Quantity: One (1) license to the software

A. Technical Specifications

- a. The software must store and process **all data on the local computer**, not requiring sending of data to any cloud-based service.
- b. The software must be able to process imu data into a kinematic model of the human over time, tracking body pose and joint angles.

IV. DELIVERABLES

<i>Description</i>	<i>Quantity or Format</i>	<i>Due Date</i>
The IMU measurement system, the EMG measurement system, and required accessories.	All required items should be shipped in a single package to the NIST Gaithersburg Campus.	1 month ARO
The software for using the sensing systems.	Should be able to be downloaded directly to the operating computer.	1 month ARO

Training Requirement	In person training session for NIST user.	TBD
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V. DELIVERY TERMS

Delivery shall be F.O.B Destination and shall occur in accordance with the delivery due dates provided in the above table.

FOB Destination means: The contractor shall pack and mark the shipment in conformance with carrier requirements, deliver the shipment in good order and condition to the point of delivery specified in the purchase order, be responsible for any loss of and/or damage to the goods occurring before receipt and acceptance of the shipment by the consignee at the delivery point specified in the purchase order; and pay all charges to the specified point of delivery. The contractor shall deliver all Line Items to:

National Institute of Standards and Technology
Shipping and Receiving
100 Bureau Drive, Building 301
Gaithersburg, MD 20899

[Benjamin Beiter](#)
202 Loading Dock

*****Due to ongoing limited site access of the NIST facilities, the Awardee SHALL confirm with the Designated Government Official and the Contracting Officer prior to ANY shipment to NIST*****

VI. TRAINING

The contractor shall conduct a training session at the NIST Gaithersburg Campus. The training shall provide a thorough demonstration of all system/solution functions, maintenance, data administration, and basic troubleshooting. The training may be completed at NIST Monday – Friday during the operating hours of (8am-5pm). The training session shall be scheduled prior to delivery of equipment to NIST, to occur within a reasonable time frame after receipt of items.

VII. INSPECTION & ACCEPTANCE

In addition to the inspection and acceptance terms articulated in 52.212-4, the Government reserves the right to perform validation testing of the capability of the delivered system to meet the minimum requirements listed above will be performed. Ideally this testing will happen coincidentally with the training session. Such tests and evaluations, if performed, shall be conducted within the environment that the system is to be operated. The Contractor has the right to be present during the tests and evaluations, if performed, at the Contractor's expense.

NIST may choose at its discretion to forego this part of acceptance testing.

A visual inspection of the equipment will be performed by the NIST SME to identify surface defects or any form of indication that any equipment was damaged during transport to NIST. The Government shall have sole discretion to require repair or replacement of damaged and/or nonconforming supplies at no cost to the Government. The Government at any time prior to acceptance shall reject the equipment due to defects and/or nonconformance. The vendor is responsible for latent defects discovered any time after final inspection. However, the extent of its liability shall be prorated over the useful life of the equipment.

Ownership of the equipment shall transfer to NIST upon acceptance by the Government.

The Government will test, inspect, and accept or reject the equipment within (insert working days it will take to inspect, accept or reject) of the receipt of the equipment unless otherwise indicated above. The Government reserves the right to conduct quality assurance testing to confirm that a given instrument(s) meets the manufacturer's and/or the Government's performance specifications. It is anticipated that the equipment will meet all manufacturer's specifications and/or the Government's performance specifications identified in the most recent operations and maintenance manual for each piece of equipment and/or in this document.

VIII. WARRANTY

The contractor shall warranty the entire system for a period of a minimum of one year after receipt of the equipment and shall be in accordance with terms in FAR 52.212-4. Warranty shall commence upon acceptance of the system by the Government and at a minimum shall include the repair or replacement of any non-consumable hardware provided by the contractor, for damage not considered to be the direct fault of the user.

IX. PAYMENT SCHEDULE (This section to be revised as needed)

Advance payment is not authorized. The Contractor must invoice in arrears according to the payment schedule. The Contractor shall be paid, in accordance with Net 30-day payment terms, upon receipt and acceptance of a proper invoice, in accordance with the following schedule:

1. 100% after installation and acceptance by the SME of fully installed system, AND
2. After successful demonstration by the instrumentation that it performs IAW the technical requirements set forth in this document.

NOTE: Partial shipments and partial invoices will not be accepted, unless otherwise requested and accepted by the Contracting Officer prior to award offer. Proposed payment schedules shall be submitted with vendor's response to the RFQ for consideration.

X. MISCELLANEOUS INFORMATION

Safety: The Contractor employee who will deliver the training must be responsible for knowing and complying with all installation safety prevention regulations. Such

regulations include, but are not limited to, general safety, fire prevention, and waste disposal.

Security: NIST is a restricted campus. An identification badge is required for access for entry into buildings and is shown to the armed Security Police when entering the campus.

Regular Business Hours. Regular business hours are Monday through Friday, (8:00 am to 5:00 pm) Eastern Time, excluding Federal holidays and NIST closures.

Identification Badges: Contractor employees must comply with NIST identification and access requirements. Each Contractor employee must wear a visible identification badge provided by the NIST Security Office.

Vehicle Registration: All Contractor employees must register their vehicles with the NIST Security Office to gain access to the campus. A valid driver's license, Government-furnished civilian ID, proof of insurance and current registration must be presented to the NIST Security Office, at which time a NIST vehicle pass will be issued. The pass must be displayed on the vehicle in accordance with NIST Security Office instructions.